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Design and Analysis of a Passive Air Conditioning System for Passenger Vehicle Cabins

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Abstract- This study explores a novel passive air conditioning approach for passenger vehicles using a combination of solar chimneys, advanced reflective glass, and humidity diffusers. The system capitalizes on environmental gradients to facilitate cooling without active refrigeration or conventional HVAC, employing blue-tinted reflective glazing for solar rejection and a solar chimney for natural venting. A diffuser, tuned to density-based air exchange, regulates moisture and temperature, while design optimizations such as window overhangs and strategic insulation further minimize heat ingress. The core strategy focuses on direct interaction between car and environment, harnessing natural forces for sustainable thermal comfort. Results demonstrate the feasibility of a car cabin system whose performance autonomously adapts to external environmental changes, establishing new possibilities for passive cooling in automotive comfort applications.

Keywords**:** Passive cooling, solar chimney, blue reflective glass, window overhang

1. Introduction

Solar energy has been used for centuries and has only improved with time, as researchers and scientists have developed processes materials to improve the quality and effectiveness of solar energy. Using solar energy, specifically passive solar energy, is a very cost efficient way to take steps towards sustainability, and as the green movement continues to develop and expand, passive solar techniques will be utilized more and more throughout the United States and in nations around the world. It is helpful to realize the history in regards to passive solar energy, and the benefits that can be gained by applying passive solar energy (Scott barber et al, 2013). Solar passive cooling is hereby presented as a pure structural and mechanical issue compared to the generally used electromechanical cooling system (Ndinechi M. C et al, 2012). A solar chimney, often referred to as thermal chimney is a way of improving the natural ventilation of buildings by using convection of air heated by passive solar energy. Solar chimneys are constructed to actively promote ventilation of unwanted heated or stale air by drawing fresh cooler air from vent sat lower levels. The exchange and movement of air-cools the building by driving heat to the outside (Mala Vr. Meht et al). Shireesha Y. et al, (2015) suggested that the temperature inside the car can achieve up to 80°C without the proposed system. They also compare the different car cooling system with portable car cooling system. The car ventilator fan uses a solar panel and battery as a source of energy to run the ventilation fan, while portable car cooling system, uses rechargeable cell as its source of energy. Advancement in the technology of passive cooling of passenger car is important because car air conditioning reduces the fuel economy. A study by Haslinda Mohamed Kamar et el, (2012) and Rugh et al, (2001) implies that the use of air conditioner to cool the passenger car significantly reduces the fuel economy by about 20% and increases emissions of nitrogen oxides (NOx) by about 80% and carbon dioxide (CO) by about 70%, although the actual numbers depend on the actual driving conditions. In this way, after going through the various literatures it was found that the cooling technique of passenger car is either done by air conditioning system or by active solar cooling. Very literatures include the passive cooling method to cool the passenger car. In the present paper, a passive cooling model of the passenger car has been introduced which have three more

attachments to the simple passenger car viz solar chimney, diffuser and window overhang. These three attachments make the car which works with the climate change.

2. Mathematical Model for experimental Analysis

Passive cooling model of a passenger car includes the three components namely solar chimney, blue reflective glass with window overhang and diffuser etc. In this section, a mathematical model has been presented for the performance analysis of experimental results which will be done in future.

2. 1. Solar Chimney

Solar chimney works on the principle of mass transfer due to density difference. Half end of chimney is colored black so as to absorb maximum solar radiation and make inside upper air of solar chimney less denser. Cabin air temperature will be below atmospheric due to proper insulation using thermocole. Hence due to density difference there is natural flow of air from cabin to environment. Solar chimney concept is the basic idea of natural circulation of air. Circulation rate depends on the chimney height and draft construction. By increasing chimney height and draft angle its efficiency can be increased. In our passenger, car chimney is placed at the roof of car so that hot air inside the cabin could circulate due to density difference.

Following equations has been used to calculate draft, pressure difference and air mass flow rate through chimney (www.engineeringtoolbox.com) **Natural draft pressure**,

$$dp = g(\rho_o - \rho_r)H \tag{1}$$

Density and temperature

$$\rho = \frac{353}{273+t} \tag{2}$$

Major and minor losses

The natural draft force will be balanced to the major and minor losses in duct, inlets and outlets. The major and minor loss in the system can be expressed as

$$dp = f\left(\frac{l}{d_h}\right)\left(\frac{\rho r v^2}{2}\right) + \Sigma\xi\,\frac{1}{2}\rho r\, v^2 \tag{3}$$

From equation (2) and (3)

$$v = \left[\frac{2g(\rho_o-\rho_r)H}{\left\{f\left(\frac{l}{d_h}\right)(\rho r)+\Sigma\xi\rho_r\right\}}\right]^{0.5} \tag{4}$$

Air flow rate through the chimney

$$q = \frac{\pi}{4}{d_h}^2 v \tag{5}$$

2. 2. Window Overhang

Here window is designed with the help of blue reflective glass in such a way that there is no direct solar radiation enters into the cabin of passenger car in summer season but the direct solar radiation in winter season must enter. In this way, Heating effect in winter season and cooling effect in summer season has been achieved.

Following formulae has been used to find the solar angle of elevation; (Thayer Watkins et al.) suggested following equations for the calculation solar angle of elevation

$$\dot{\eta} = 90° - \cos^{-1}\{\cos\phi\cos\psi\cos\theta + \sin\phi\sin\psi\}$$

where , $\psi = 23.5°\sin\left\{\frac{360T}{365.25}\right\}$

$$\theta = 360°\left(\frac{W}{24}\right) \tag{6}$$

Calculation of length of window overhang:-

$$x = H/tan \tag{7}$$

Heat transmitted through south facing glass

Heat transmitted through south facing glass can be calculated by using following equation mentioned in book (CP Arora refrigeration air conditioning and heat transfer, Tata McGraw hill education private limited, New Delhi)

$$Q = (A_{sun}\tau_D I_D + A\tau_d I_d) + \frac{A_{sun}a_D I_D + Aa_d I_d}{1+\frac{f_o}{f_i}} + UA(t_o - t_i) \quad (8)$$

When the glass window is not directly exposed to solar radiation then
$A_{sun=\,D} = I_D = 0$,

3. Conclusion

Following conclusions has been drawn from the present work which is as follows-

- Present model of a passenger car works with climate change. It has three attachments (diffuser, solar chimney and blue reflective glasses) which make our house different from general house.
- Diffuser helps to flow cold air from diffuser tank to house. Evaporation process at the surface helps air at diffuser inlet end cool that is why car is cooled by evaporative cooling process.
- Chimney draws hot air inside the room and sends it to surroundings. This is the natural circulation process and happens due to density difference of air. For creating difference in density chimney is blackened and for creating drought it has been tapered.
- Role of glasses in passenger car are to make cabin lighted and to control and utilize solar radiations.
- On the whole it can be said that there are three processes which make general house to solar housefirst one insertion of cold air, second one rejection of hot air and third one is protection from solar radiation in summer season.

Numerical model for the analysis of present work has been presented in this paper but experimentation has still to be performed for its verification.

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Nomenclature
ρ_c = Density of air inside the cabin,
ρ_o= Density of air outside the cabin,
H= Height of chimney, g=
Acceleration due to gravity, t_c =
Temperature inside the cabin, t_o=
Temperature outside the cabin,
dp= Pressure difference between inside and outside of the cabin,
f= Darcy-weisbac friction coefficient l= Length of duct or pipe
(m) d_h= Hydrolic diameter (m) $\Sigma\xi$= Minor loss coefficient v=
Velocity of air through chimney, t= Actual temperature

$$d_h = \frac{d_i + d_o}{2}$$

d_i= Dia of base of
chimney d_o= Dia of top of
chimney d_h= Hydrolic mean
diameter
= Solar angle of elevation with horizontal,
= Angle of latitude,
= The angle which the sun's ray makes with the x-axis,
= Longitude angle corresponding to time t of hours past noon angle (hour angle),

T= The days past the spring equinox
w= The time of hour past noon x=
Length of overhang
A_{sun}= Glass area directly exposed to the sun
$_D$= Transmitivity of glass for direct solar radiation I_D=
Intensity of direct solar radiation
a_D= Absorptivity of glass corresponding to direct solar radiation
$_d$= Transmitivity of glass for diffused solar radiation I_d=
Intensity of diffused solar radiation
a_d= Absorptivity of glass corresponding to diffused solar radiation
t_o= Outside temperature t_i= Room temperature f_o= Outside film-coefficient of heat transfer through glass f_i= Inside film-coefficient of heat transfer through glass
U= Overall coefficient of heat transfer
A= Total area of glass
C= Dimensionless coefficient for the month of may.
I_n= Direct solar radiation normal to the sun's ray .
F_{ss}= Angle factor between surface and sky
F_{sg}=Angle factor between surface and ground
L=Air mass λ=angle of tilt surface
with vertical

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